

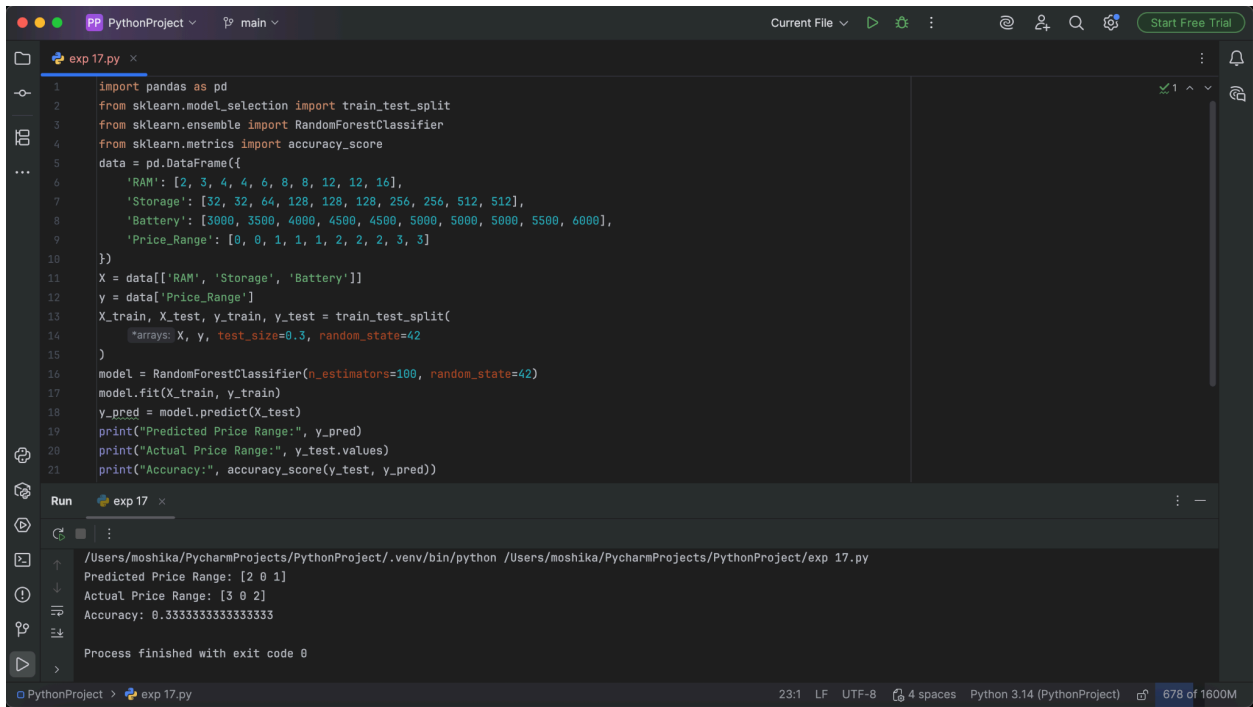
EXPERIMENT - 17

Aim: To predict the price range of mobile phones using a suitable machine learning algorithm.

Procedure:

1. Load the mobile price dataset.
2. Preprocess the dataset.
3. Select relevant features such as RAM, battery, storage, and screen size.
4. Separate input features and price range.
5. Split the dataset into training and testing sets.
6. Train a classification model.
7. Predict the price range for test data.
8. Calculate the accuracy.
9. Display the prediction results.

Output:



```
1 import pandas as pd
2 from sklearn.model_selection import train_test_split
3 from sklearn.ensemble import RandomForestClassifier
4 from sklearn.metrics import accuracy_score
5 data = pd.DataFrame({
6     'RAM': [2, 3, 4, 4, 6, 8, 8, 12, 12, 16],
7     'Storage': [32, 32, 64, 128, 128, 128, 256, 256, 512, 512],
8     'Battery': [3000, 3500, 4000, 4500, 4500, 5000, 5000, 5000, 5500, 6000],
9     'Price_Range': [0, 0, 1, 1, 1, 1, 2, 2, 2, 3]
10 })
11 X = data[['RAM', 'Storage', 'Battery']]
12 y = data['Price_Range']
13 X_train, X_test, y_train, y_test = train_test_split(
14     X, y, test_size=0.3, random_state=42
15 )
16 model = RandomForestClassifier(n_estimators=100, random_state=42)
17 model.fit(X_train, y_train)
18 y_pred = model.predict(X_test)
19 print("Predicted Price Range:", y_pred)
20 print("Actual Price Range:", y_test.values)
21 print("Accuracy:", accuracy_score(y_test, y_pred))
```

Run exp 17

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 17.py
Predicted Price Range: [2 0 1]
Actual Price Range: [3 0 2]
Accuracy: 0.3333333333333333
Process finished with exit code 0
```

Result: The machine learning model successfully predicted the appropriate mobile price range.